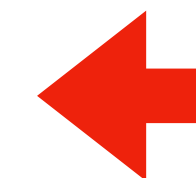
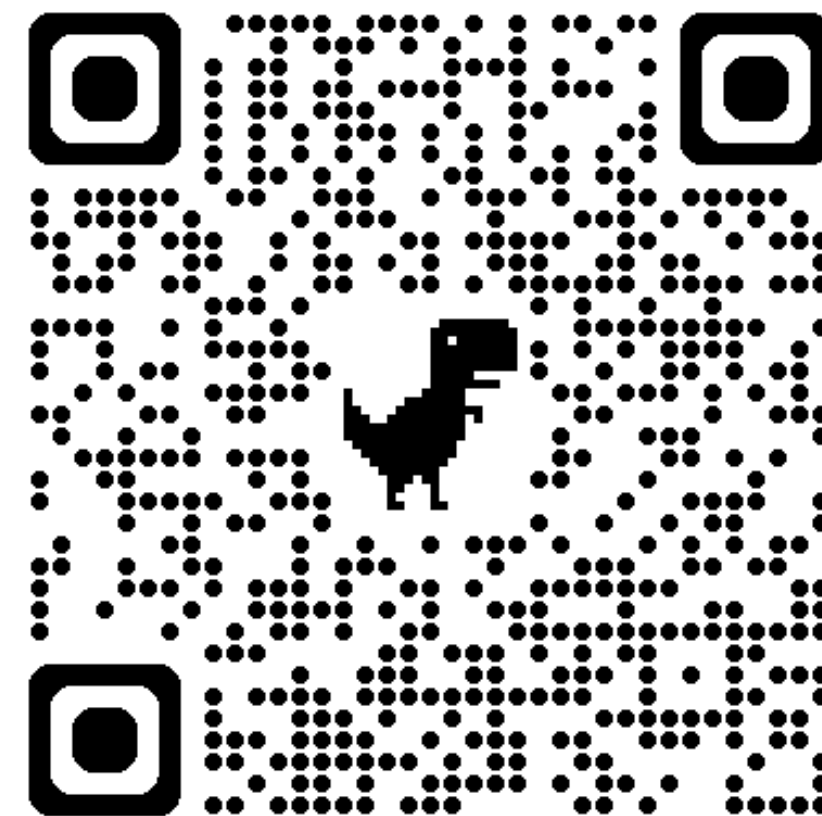


Academic Paper Writing in Artificial Intelligence

Paper Figures: Method Overview

Instructor: Ying-Jia Lin
October 1st, 2026



Couse Website
Download slide here!

Outline

- Last week's summary (15 min)
- Introduction to "captions" (圖說) (10 min)
- How to write the "captions" (20 min)
- **Your turn to revise and write (45 min, including break)**
- How to make a descent method overview figure (20 min)
- **Your turn to revise and write (40 min)**

Last week's summary

- Your **Methods part** (next week) should be separate from Evaluation metrics.
 - We propose a training framework about xxx.
 - Before evaluations, the classifier was created using the training set.
 - ...

New Policy

- Tracking changes is extremely important for practicing English writing.
- Please upload next assignments using **.docx (Microsoft Word)** instead of LaTeX PDF. No regulations for the filenames.
- I apologize for switching different tools and will briefly introduce the reference management software (Zotero) next week. LaTeX won't be used in the following classes (still welcome any questions).

Figure Caption

What is a caption? (1/2)

Jin, Qiao, et al. "Medcpt: Contrastive pre-trained transformers with large-scale pubmed search logs for zero-shot biomedical information retrieval." Bioinformatics 39.11 (2023): btad651.

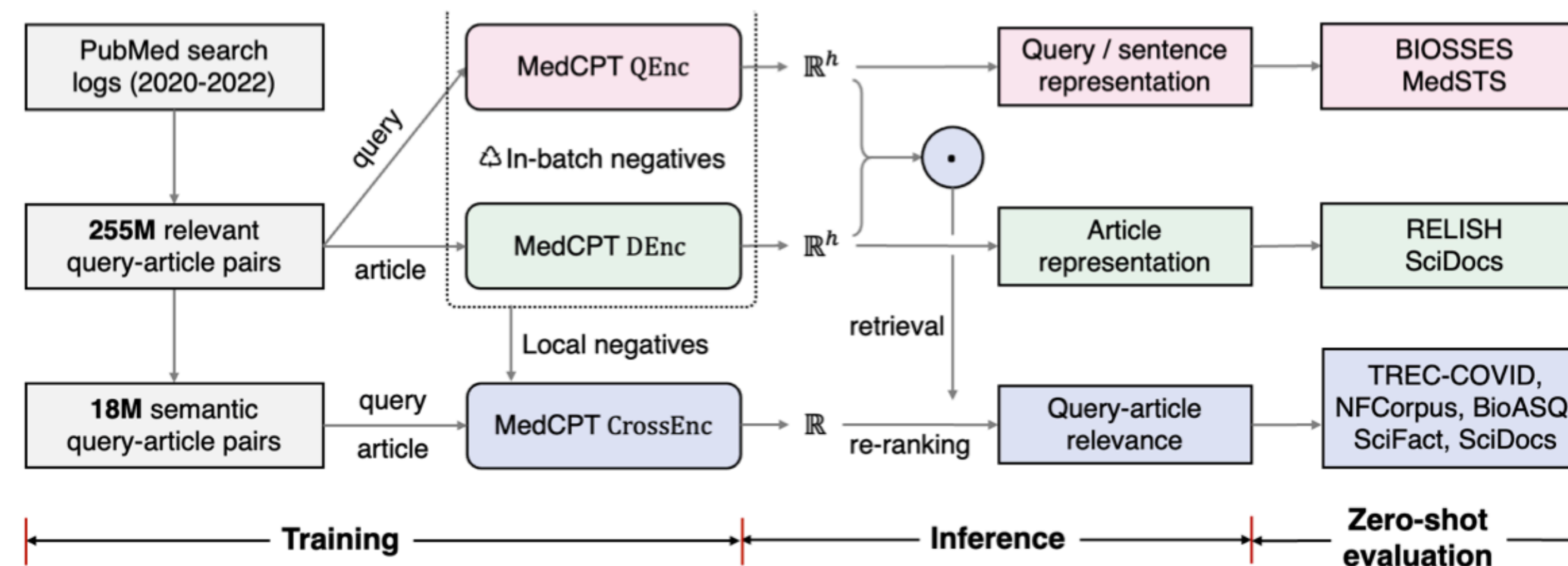
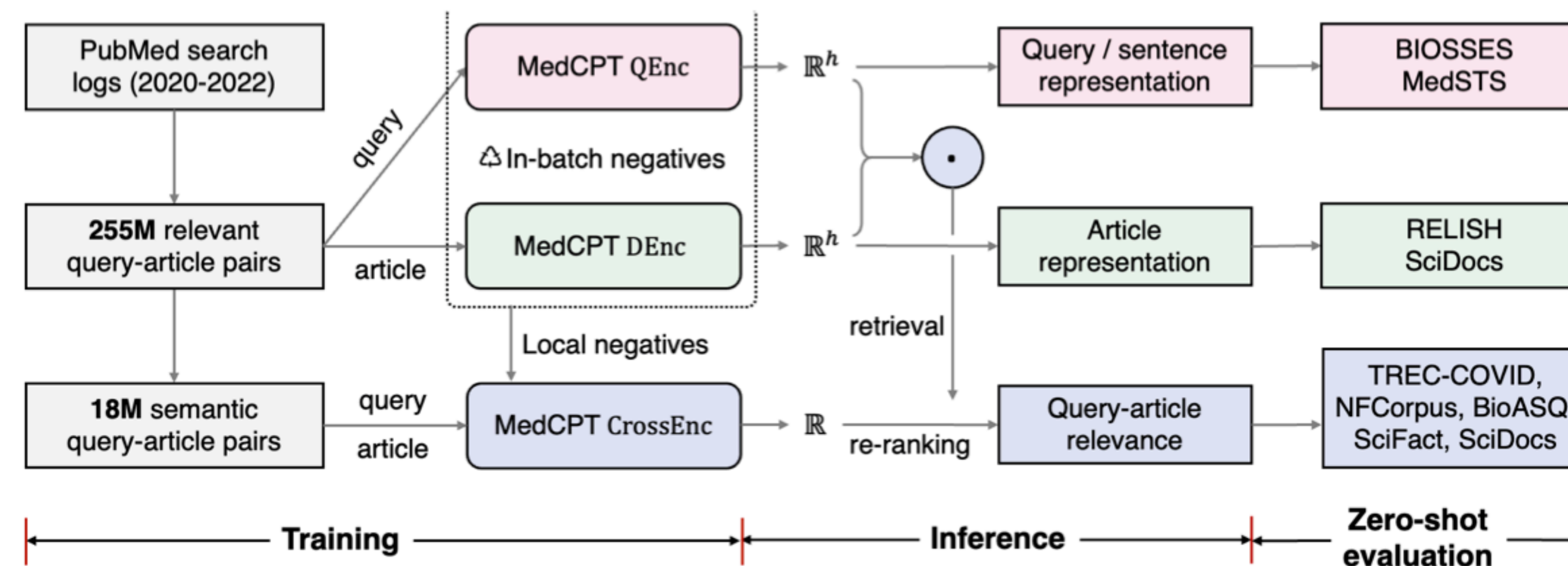


Figure 1. A high-level overview of this work. MedCPT contains a query encoder (QEnc), a document encoder (DEnc), and a cross-encoder (CrossEnc). The query encoder and document encoder compose of the MedCPT retriever, which is contrastively trained by 255M query-article pairs and in-batch negatives from PubMed logs. The cross-encoder is the MedCPT re-ranker, and is contrastively trained by 18M non-keyword query-article pairs and local negatives retrieved from the MedCPT retriever. MedCPT achieves state-of-the-art performance on various biomedical information retrieval tasks under zero-shot settings, including query-article retrieval, sentence representation, and article representation.

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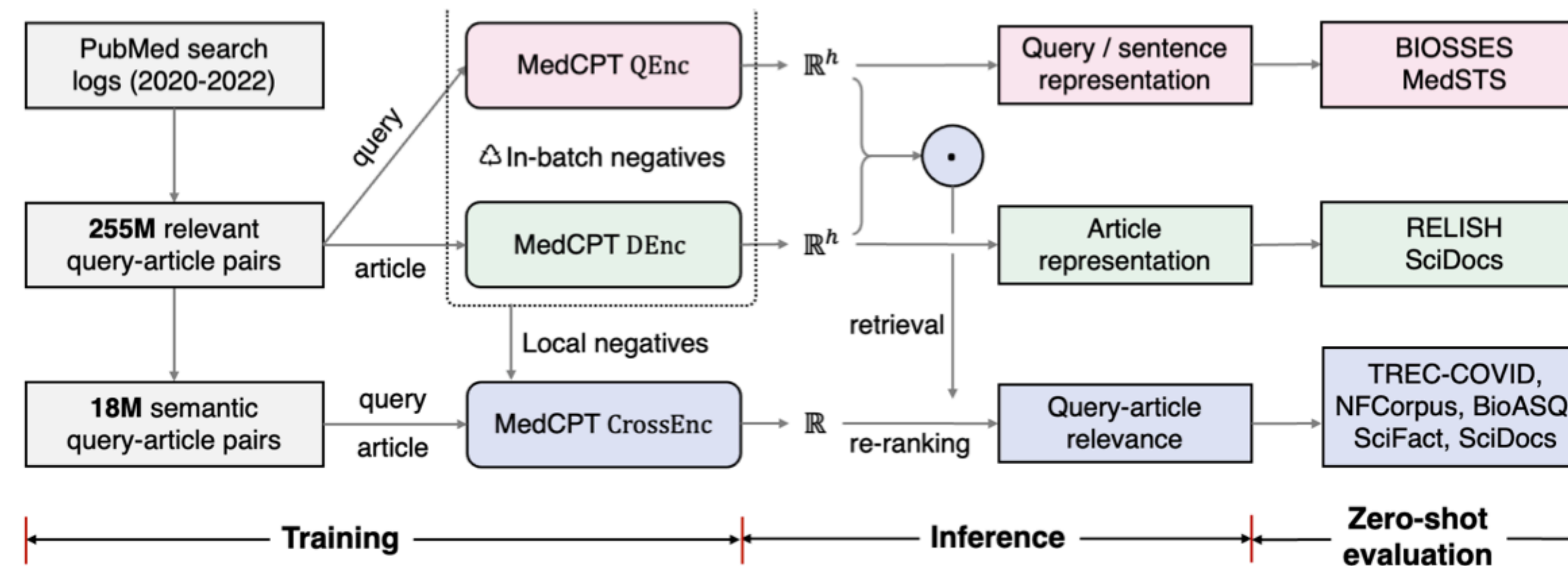


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- A caption is the only text that travels with the figure.

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- **✗** Figure 1: Model architecture.

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What it is.

How to read it.

What to notice. (Optional)

Legend vs. Caption

Legend vs. Caption



- Legends help readers **easily** read the symbols.
- Captions tell readers what the figure is about.

Legend vs. Caption



- Legends help readers **easily** read the symbols.
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Legend:

— Adam

--- SGD

Figure 2. Training loss curves of Adam and SGD on the validation set.



Caption

Three steps to write a (good) caption

1. What it is — a noun phrase, not a sentence.

The proposed training framework for unsupervised Chinese word segmentation.

2. How to read it — name the things that readers have to work out.

Green boxes are pre-trained models; orange boxes are the steps we introduce. A closed lock means the parameters are frozen.

3. What to notice — tell the hidden things or key takeaways (optional).

The classifier predicts a 0/1 boundary label for every character.

Caption Example 1 (not quite good)

Li, Hsiu-Wen, et al. "Improved unsupervised Chinese word segmentation using pre-trained knowledge and pseudo-labeling transfer." EMNLP 2023.

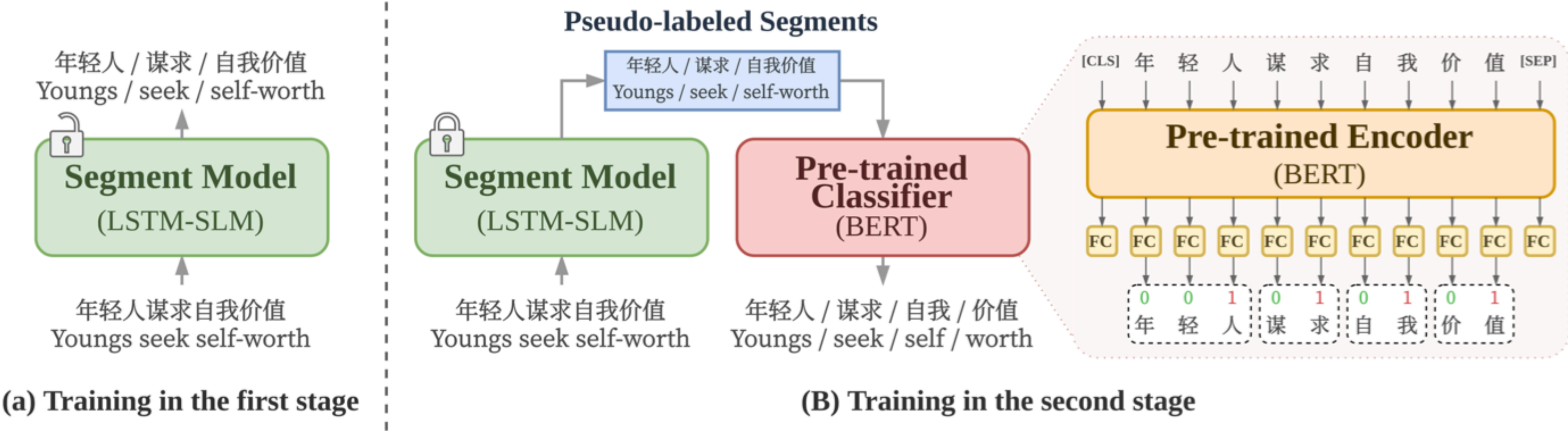


Figure 1: The proposed training framework for unsupervised Chinese word segmentation.

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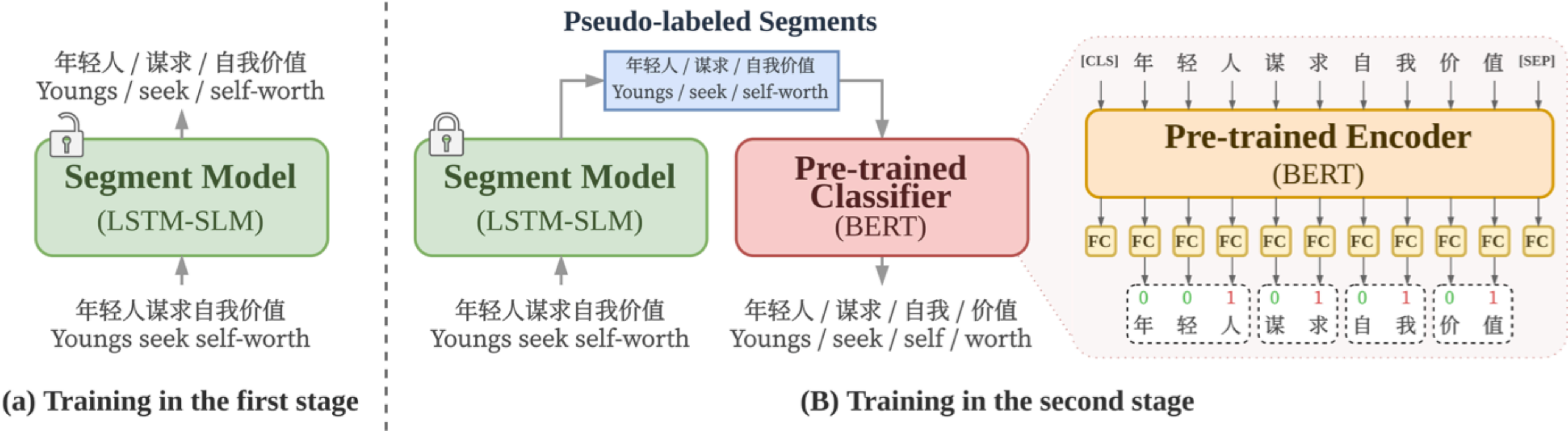


Figure 1: The proposed training framework for unsupervised Chinese word segmentation. **Step1**

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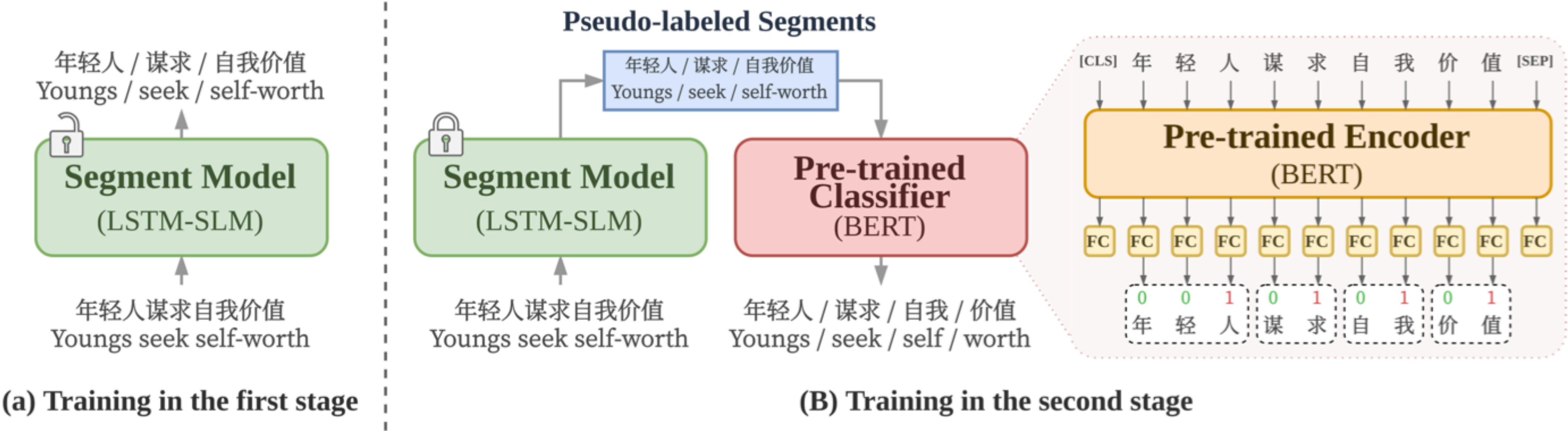


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A closed lock means the parameters are frozen. FC indicates fully-connected layers.

Step2

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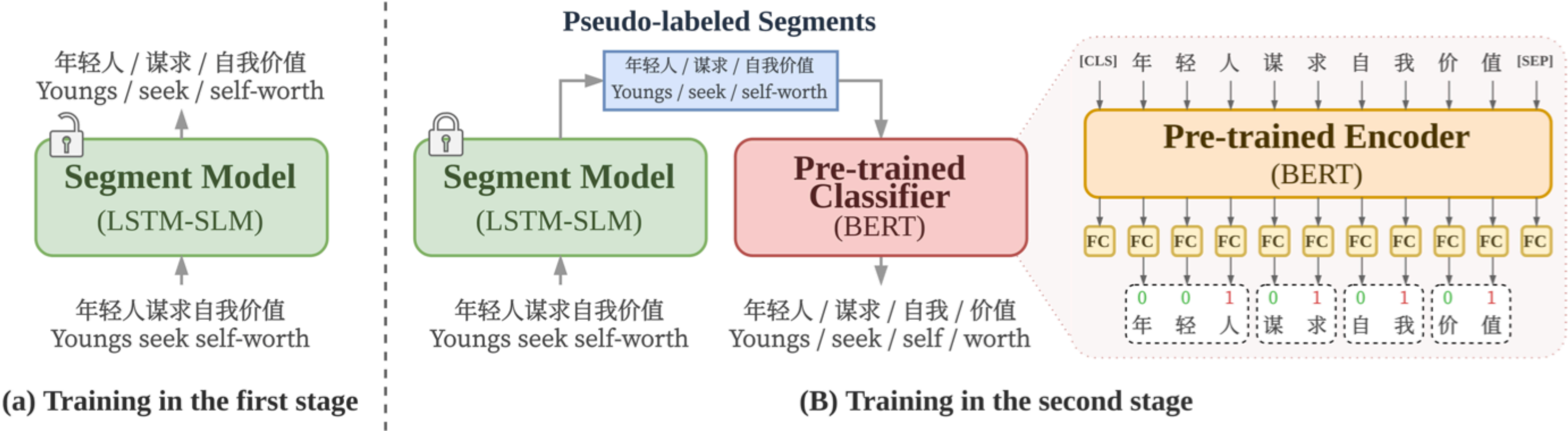


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The fine-tuned classifier predicts a 0/1 boundary label for every character.

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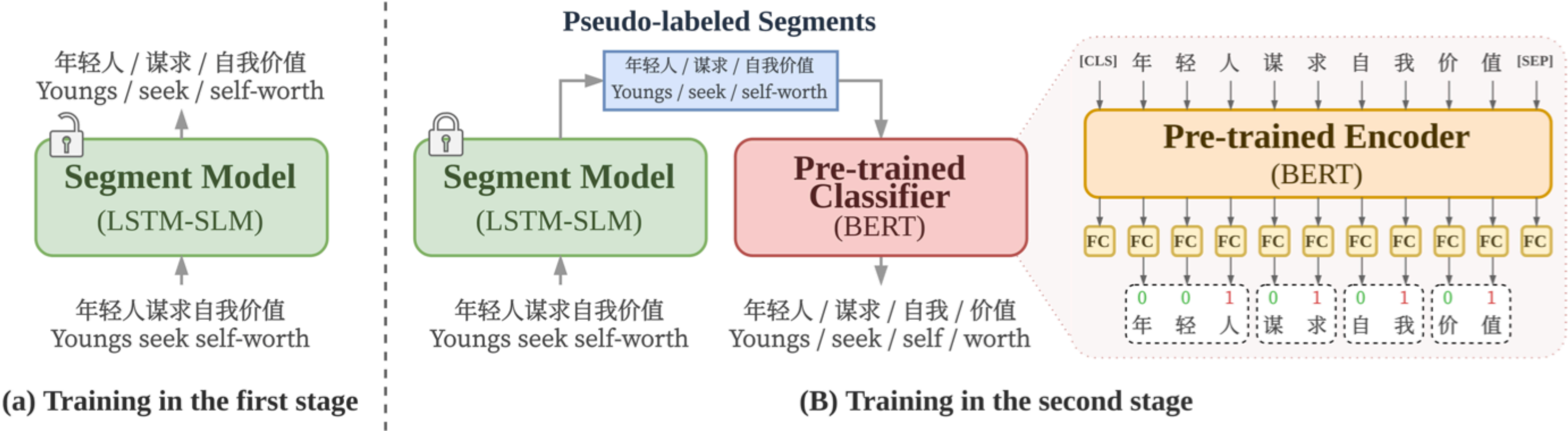


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Step3

(For course demo. We usually don't add a line break for each sentence.)

Caption Example 2 (MedCPT again)

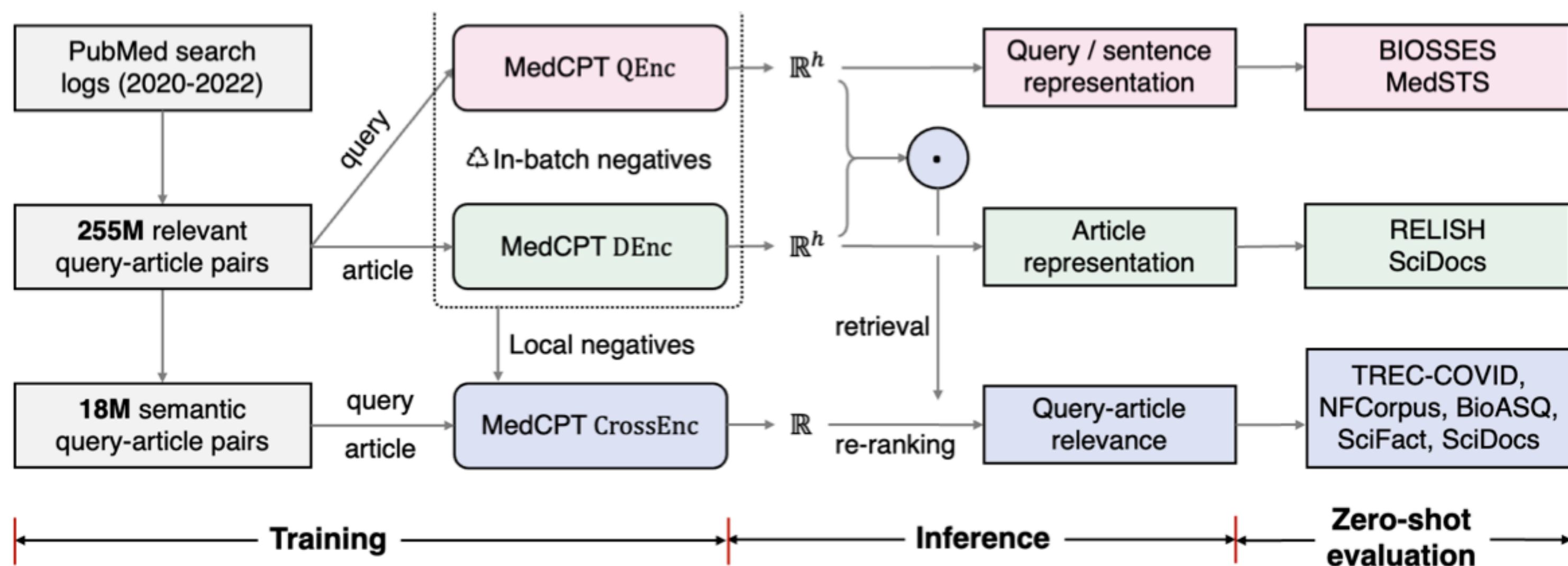


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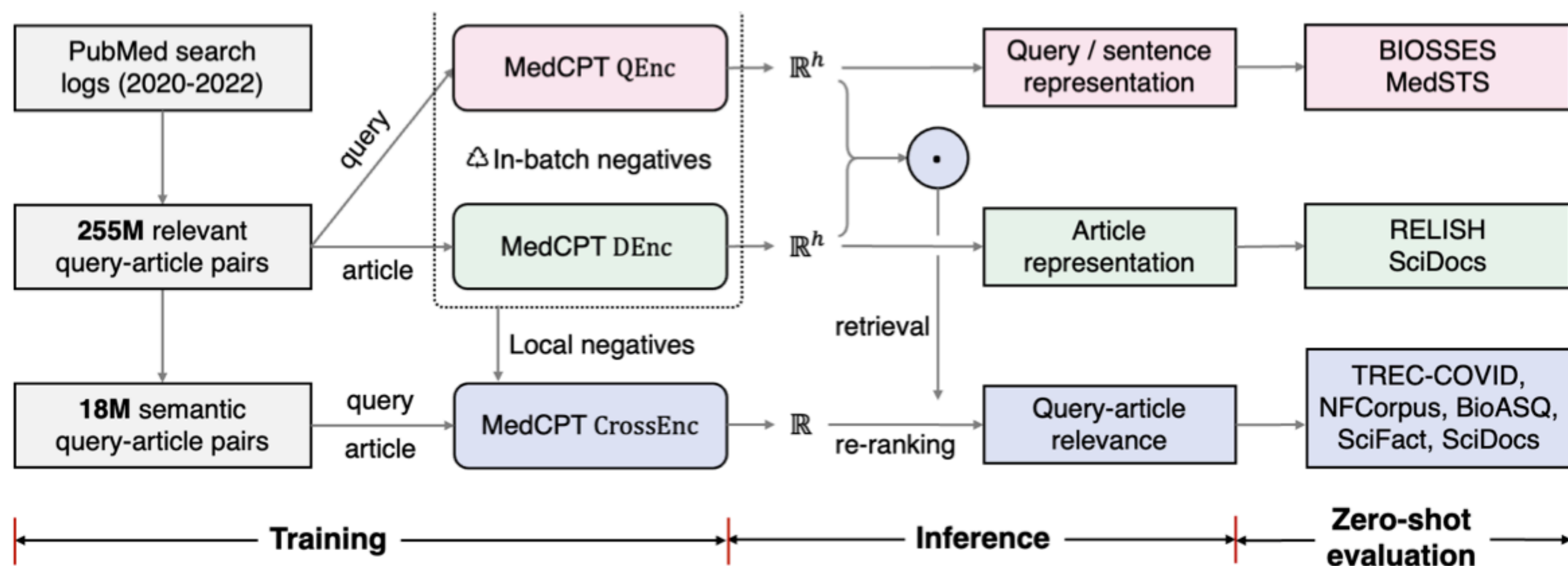


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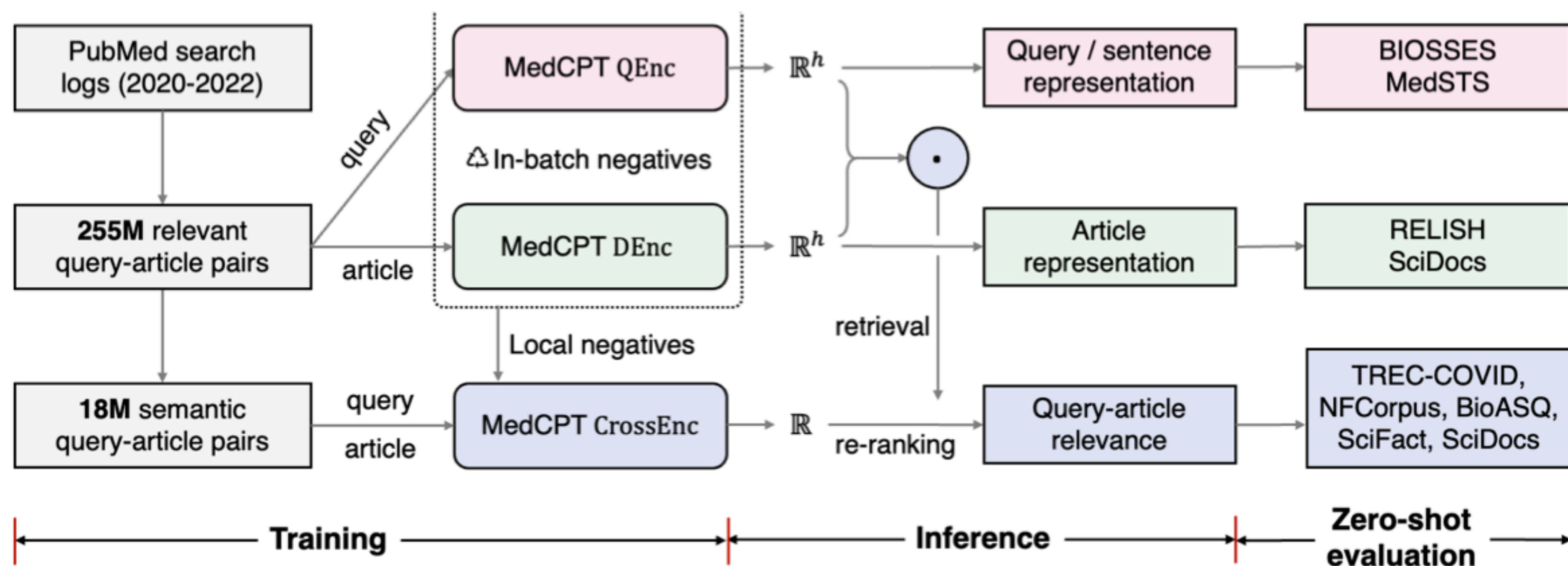


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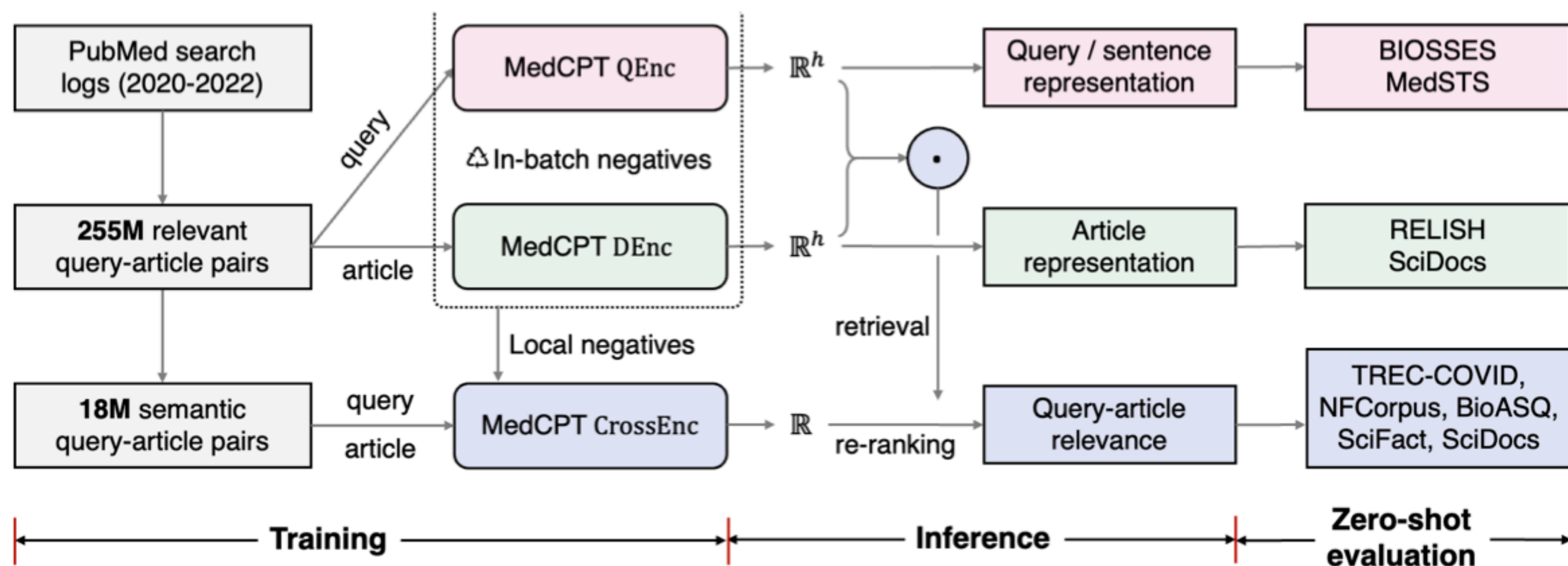


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Step1

Step2

Step3

The standalone test (self-contained)

- 💡 A reader who reads only the figure and its caption should understand your method.
- ⚠️ This is not a nice extra. It is how figures are actually read:
 - *reviewers skim the figures* before reading the text
 - *readers first meet Figure 1 on page 2*, before the Method section on page 4
- **If your caption only makes sense after reading Section 3, the caption has failed.**

Caption Example 3 (not quite good)

Lee, Jinhyuk, et al. "BioBERT: a pre-trained biomedical language representation model for biomedical text mining." *Bioinformatics* 36.4 (2020): 1234-1240.

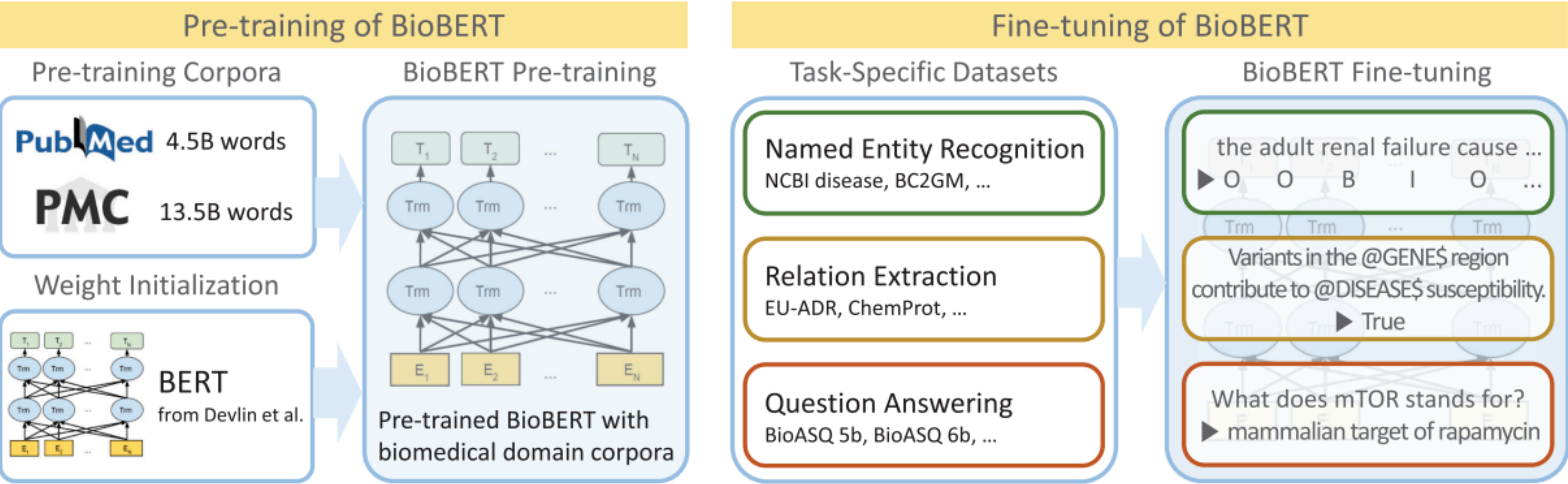
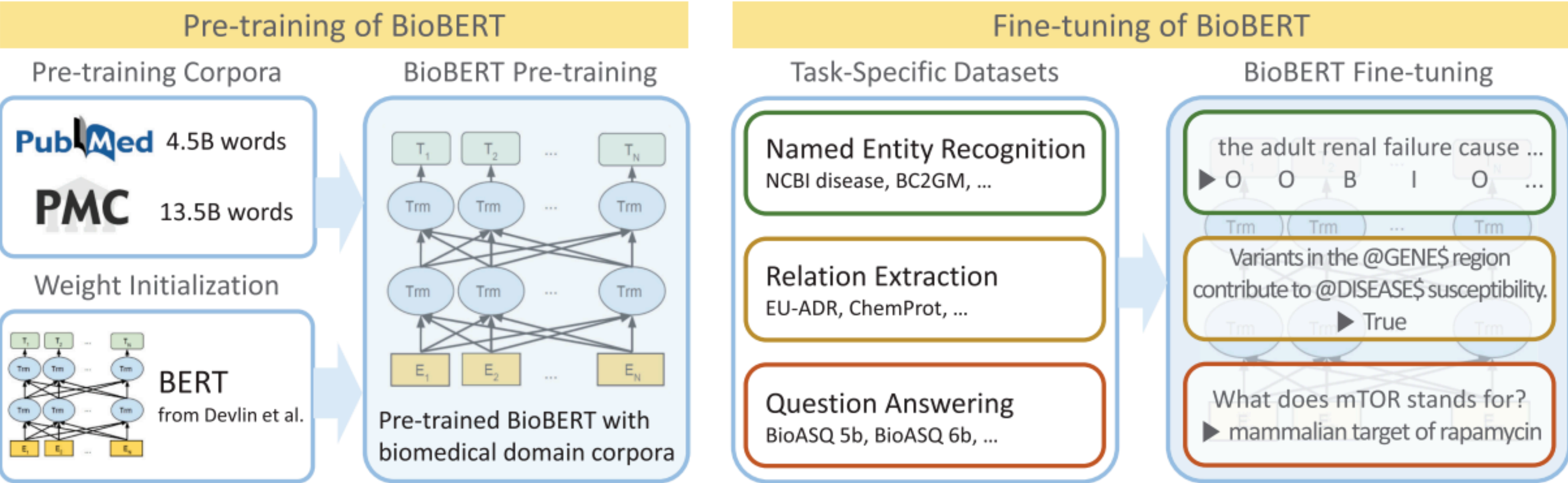


Fig. 1. Overview of the pre-training and fine-tuning of BioBERT

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➡ Fig. 1. Overview of the pre-training and fine-tuning of BioBERT

A good figure with a one-line caption is still a broken figure.
Though the figure is a strong one, its caption does not guide readers to understand it.

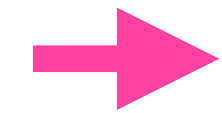
Your Turn

How to make an overview figure?

Three kinds of method figure (today: middle)

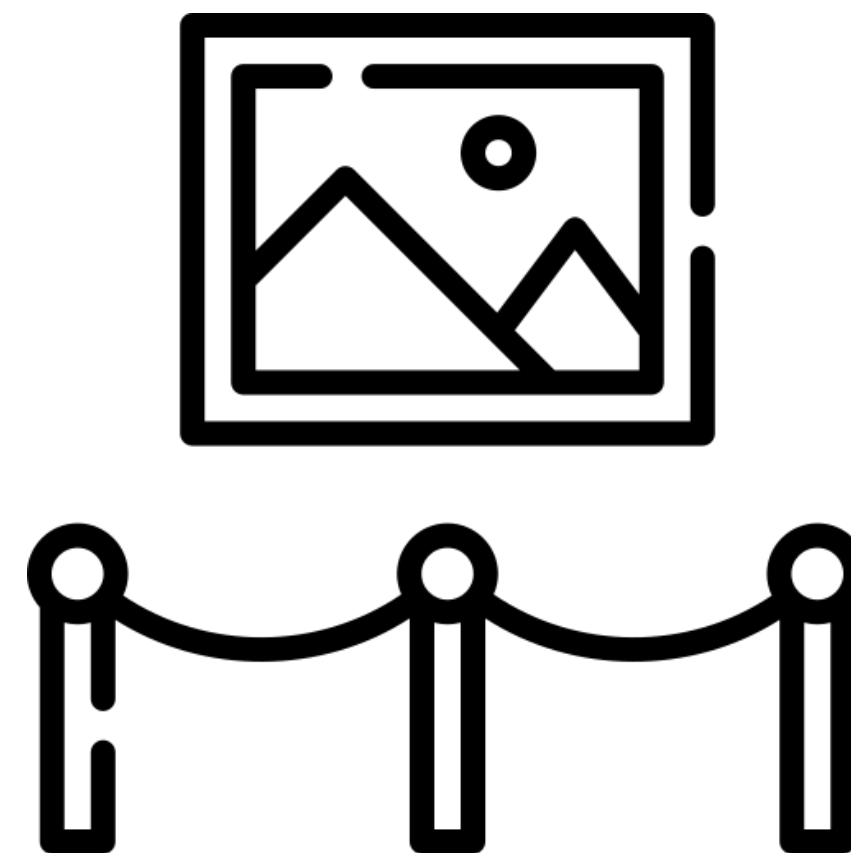
Kind	Use it when
Task Illustration	your method may be simple, but the task is unfamiliar
Pipeline	your work has stages that run in order
Architecture	you changed the model itself

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Gallery First



Example: Task Illustration

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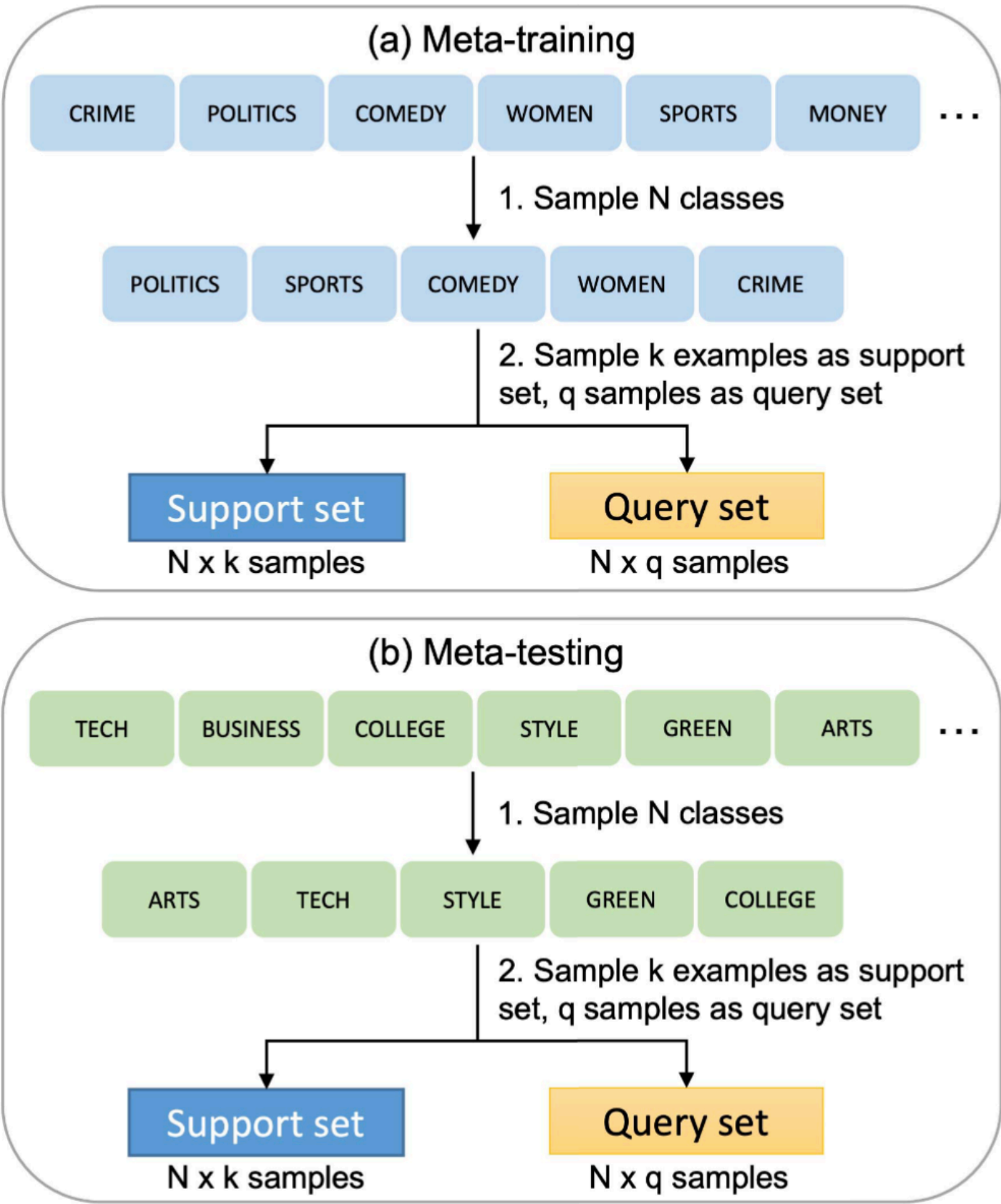


Fig. 2. Episode generation during meta-training and meta-testing. The classes in the meta-training and meta-testing phases are disjoint.

Evaluation



The driver offered a ride to the teacher but he refused the offer. who is 'he'?



Since the driver is the one making the offer, **it follows that 'he' in the question refers to the driver. The teacher's refusal** of the offer does not affect who is being referred to in the initial action.



Isn't it contradictory? The reasoning says '**he is the driver**' but states it is the teacher who refused the offer. So '**he**' should refer to the teacher.

Detection



Do you think the reasoning above is self-contradictory given the question and prediction?



I couldn't see the contradiction of the reasoning. Looks fine to me.

Figure 1: An example for self-contradictory reasoning and detection by LLMs. LLMs fail to generate consistent reasoning and are poor at detecting the self-contradiction.

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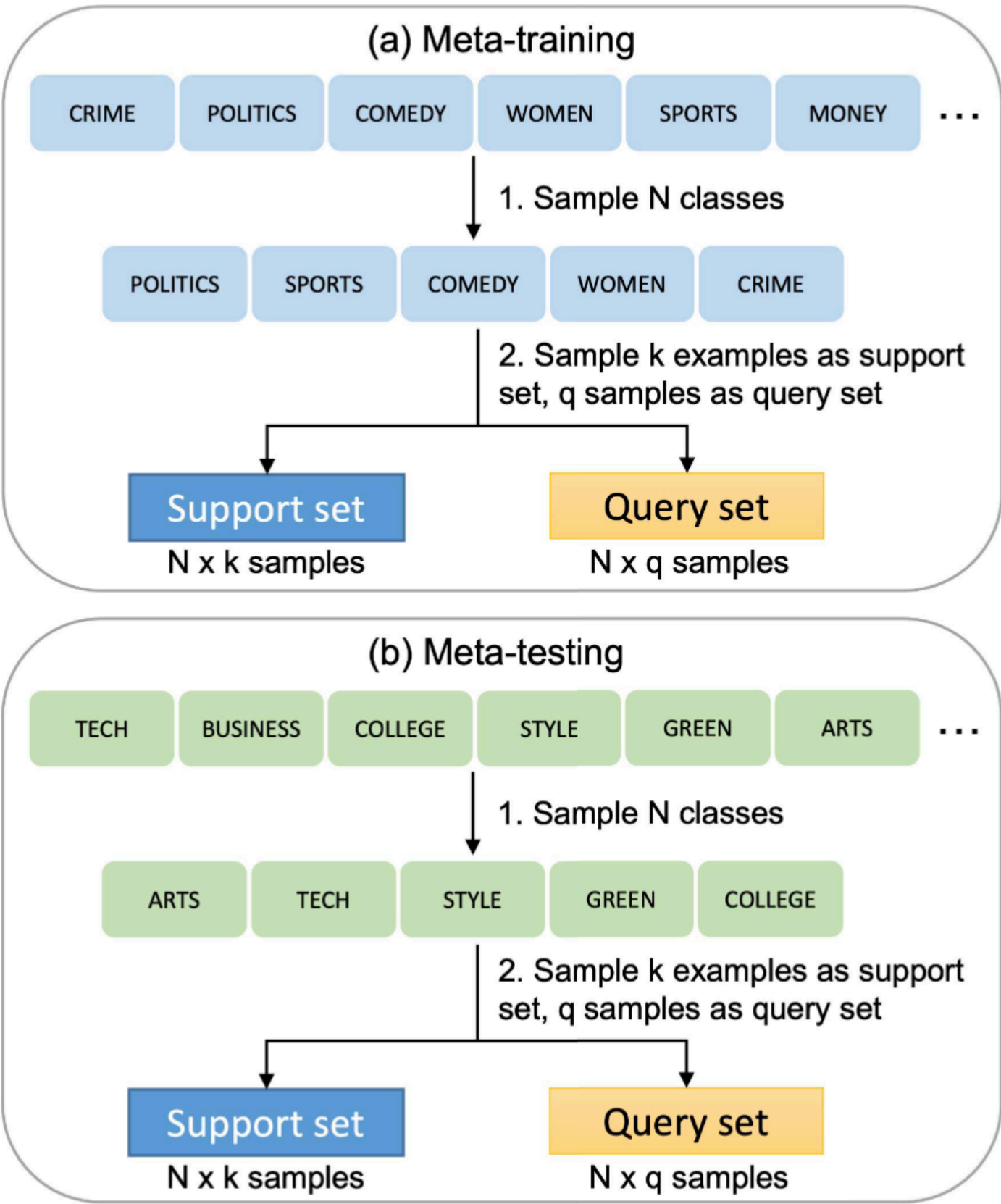


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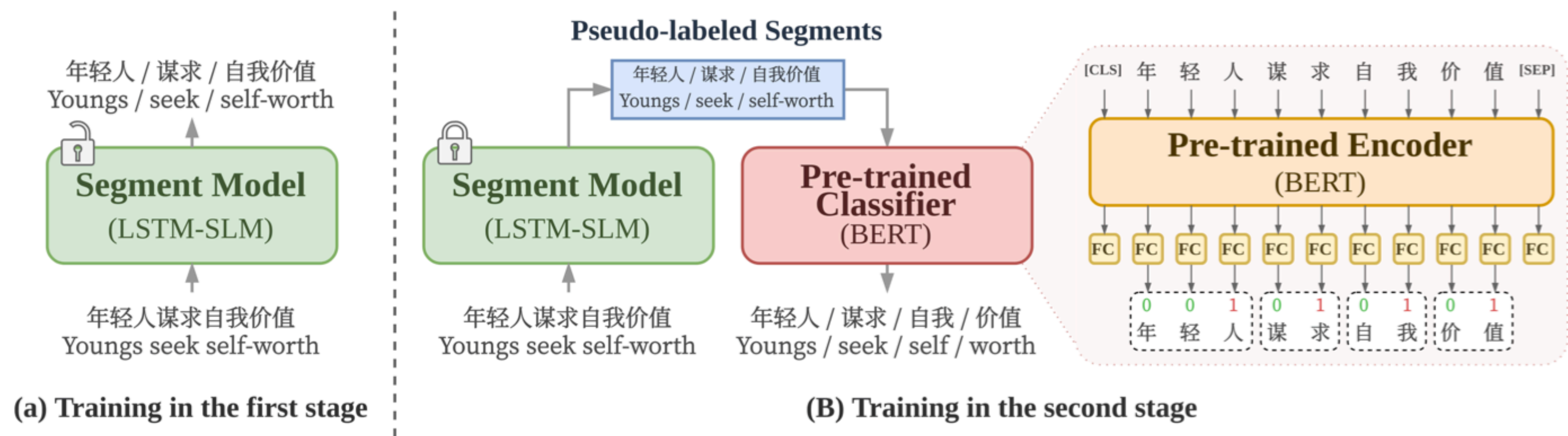


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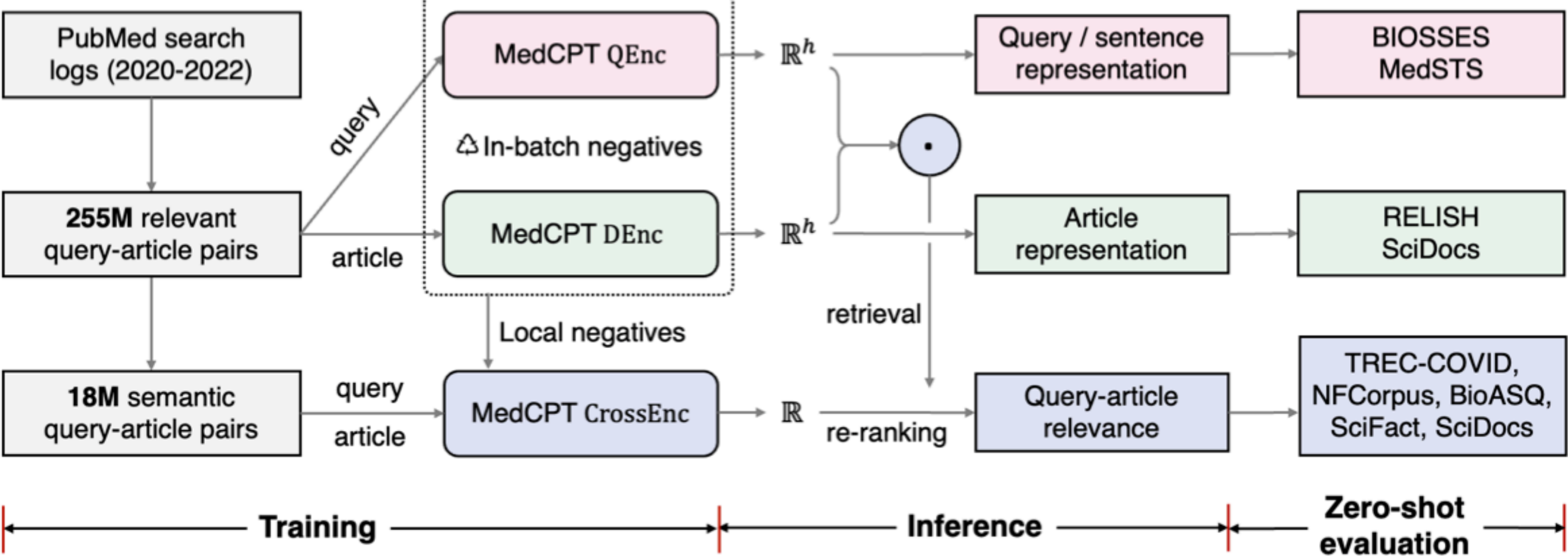


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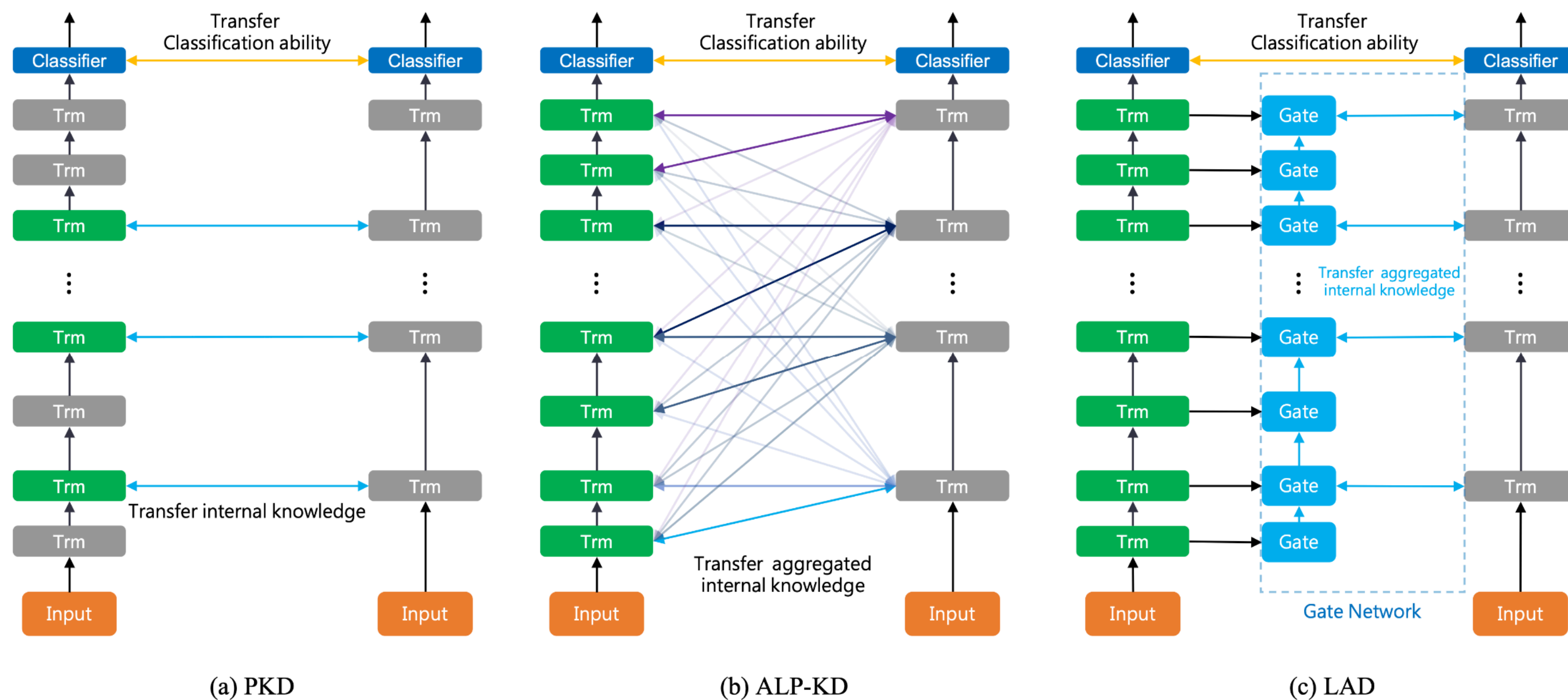


Figure 1. Differences in model structures of three distillation frameworks. Trm means a Transformer layer. In each subplot, the left shows the teacher model, and the right indicates the student model. (a) Patient knowledge distillation (PKD) [14]; (b) Attention-Based Layer Projection for Knowledge Distillation (ALP-KD) [15]; (c) Our proposed method.

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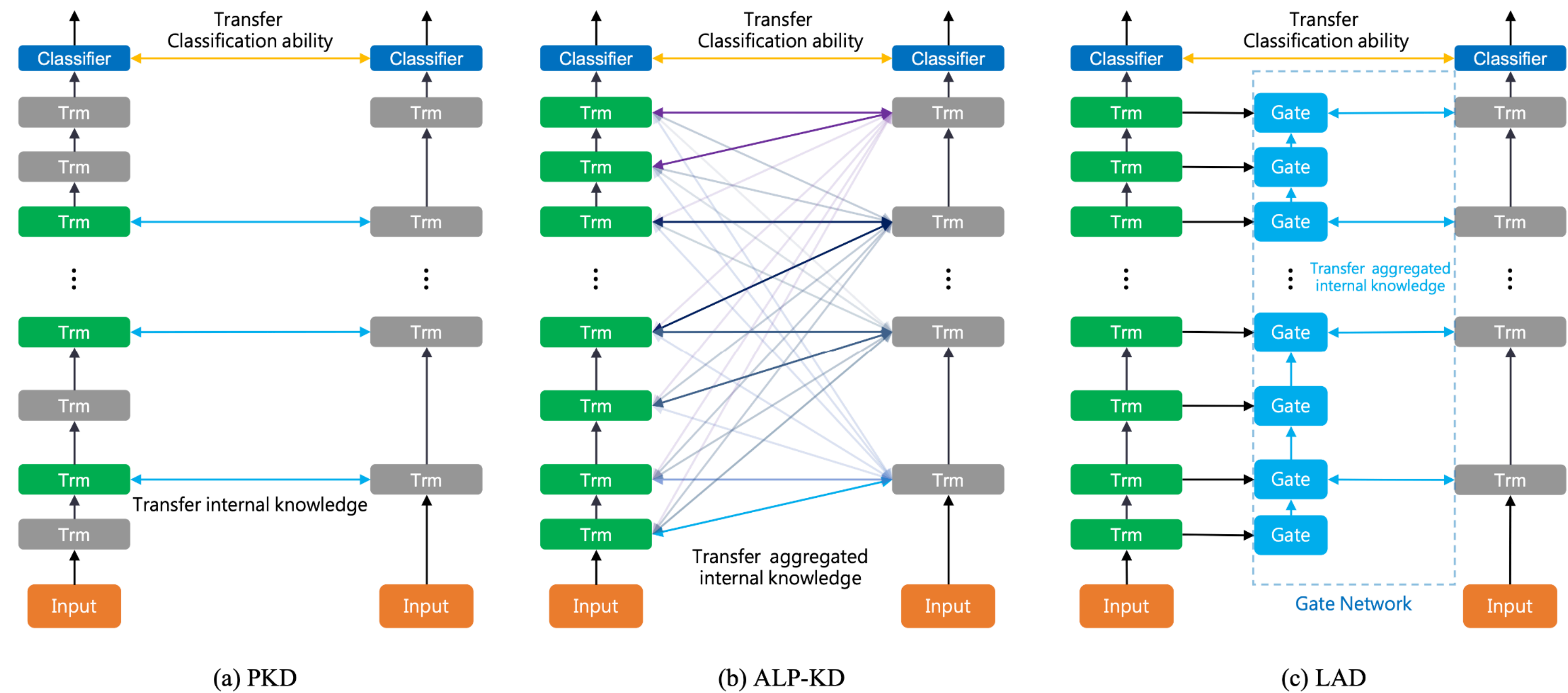


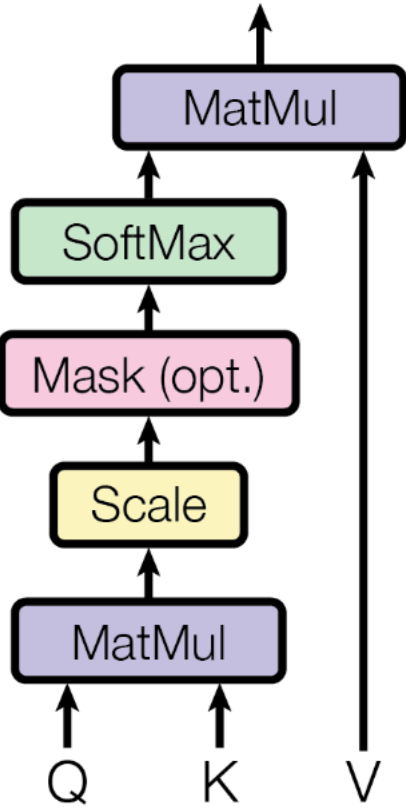
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Scaled Dot-Product Attention



Multi-Head Attention

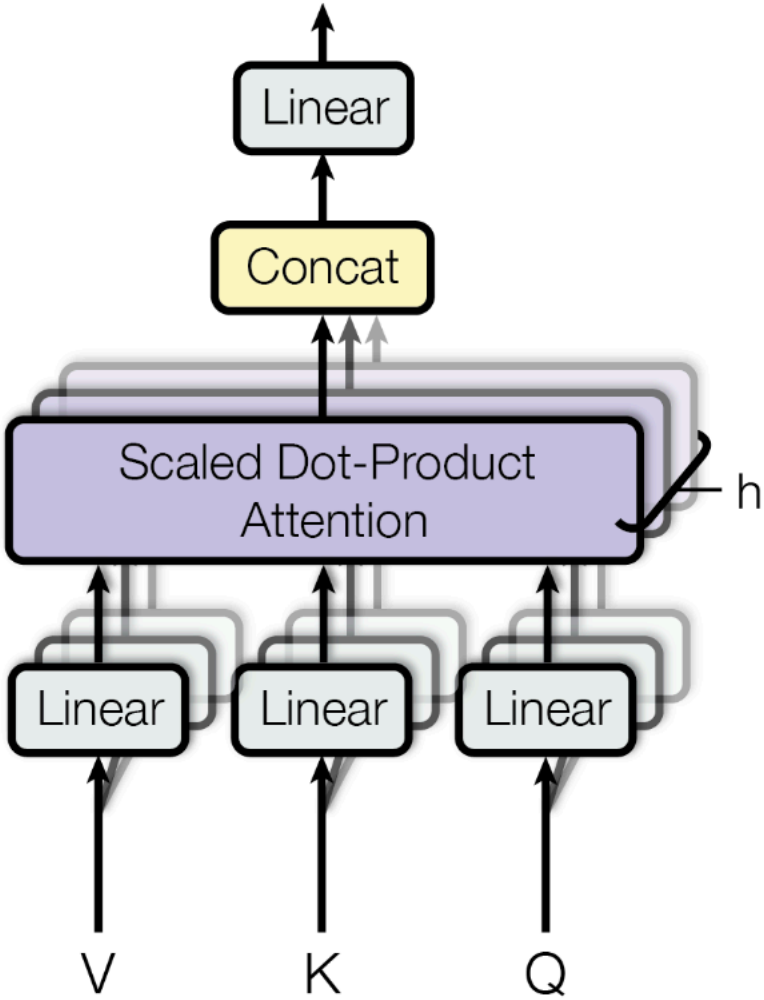


Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.

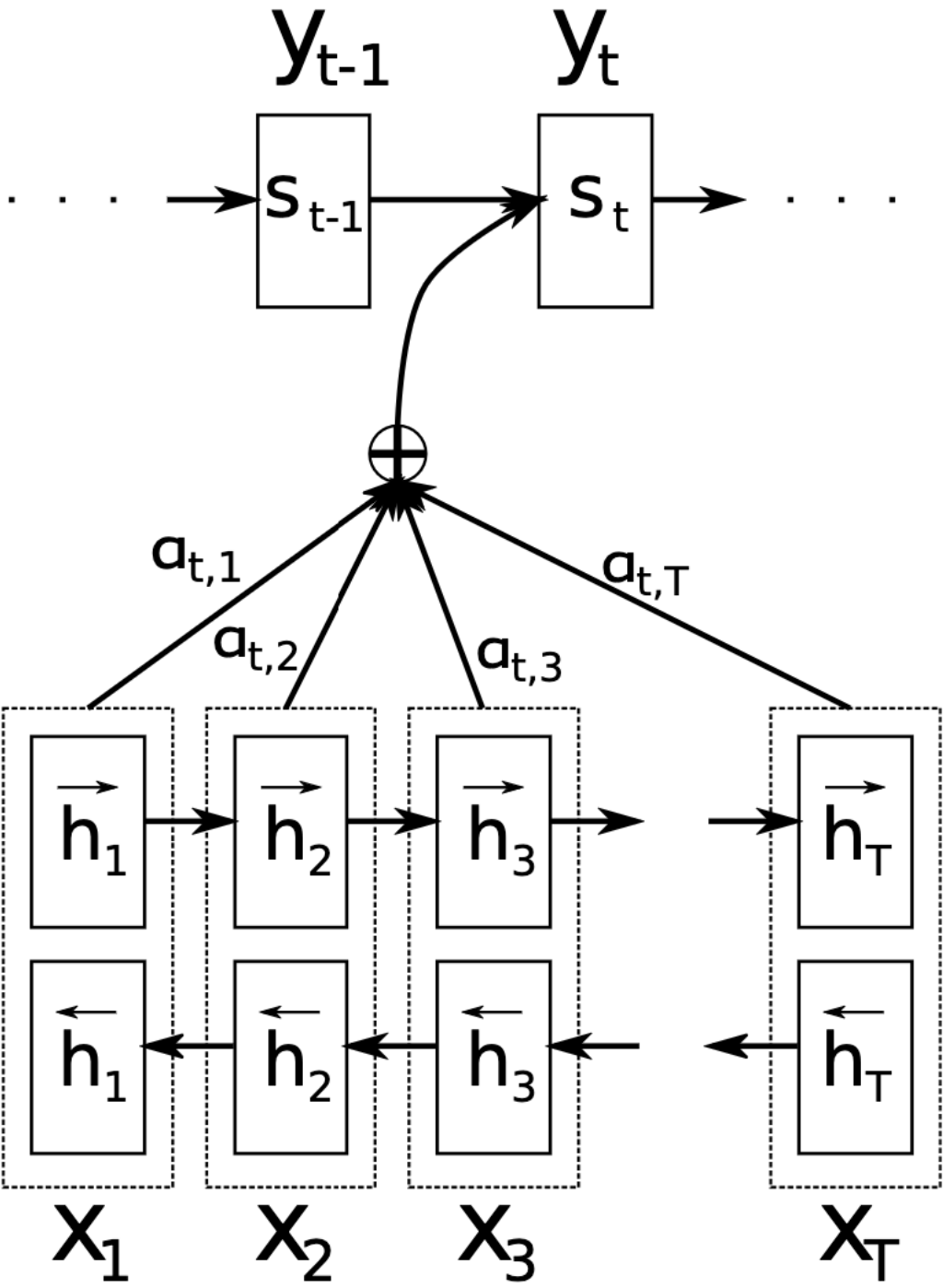


Figure 1: The graphical illustration of the proposed model trying to generate the t -th target word y_t given a source sentence $(x_1, x_2, \dots, x_T)$.

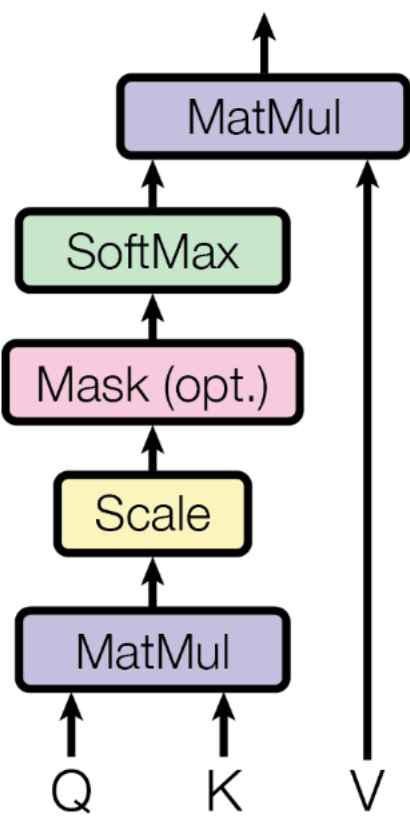
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Scaled Dot-Product Attention



Multi-Head Attention

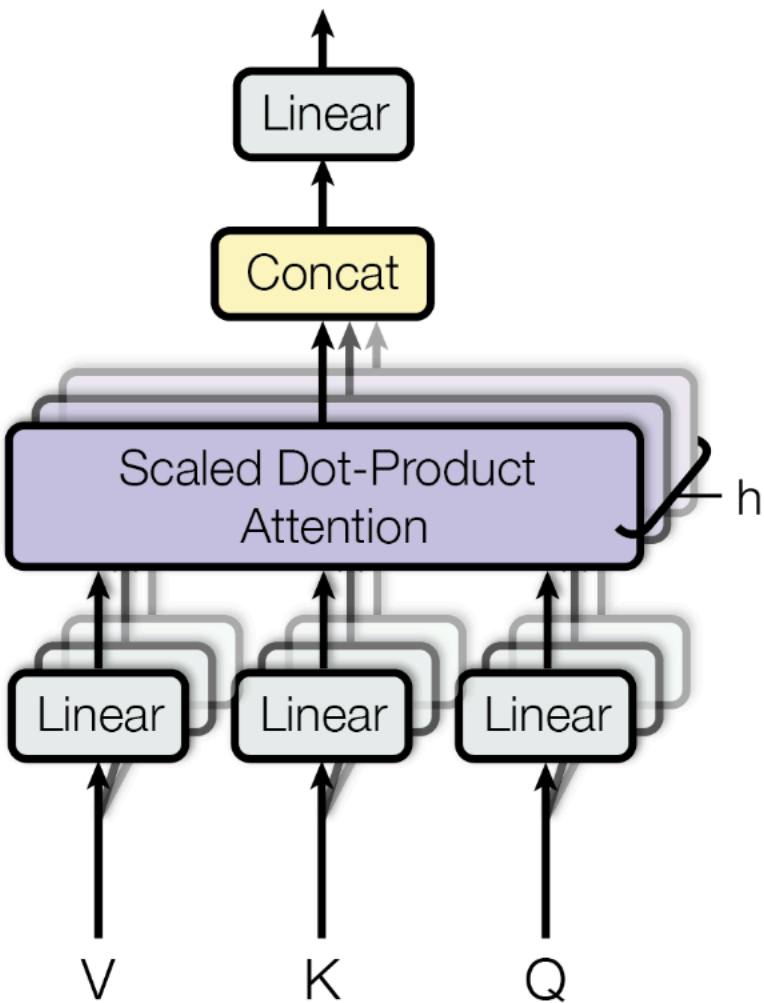


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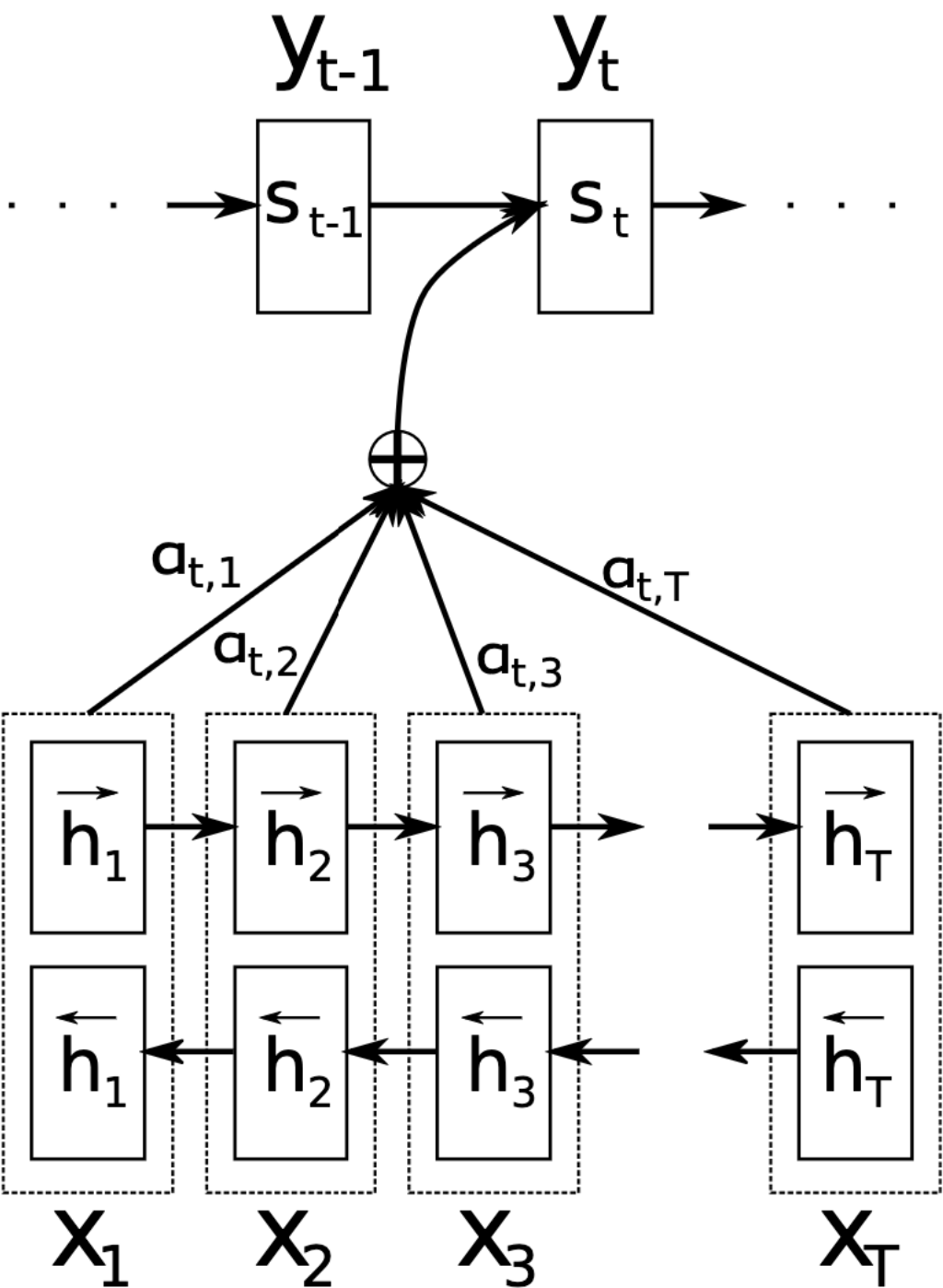


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Key Findings

1. The overview figure **follows your contribution**, not your taste.
 - unfamiliar task → illustration | stages in order → pipeline | model changed → architecture
2. They all read **in one direction**.
 - Input at one end, output at the other. None of them makes you hunt for the start.
3. **Real names, real numbers (more information)**
 - "255M query–article pairs", "TREC-COVID", "Nx" — not "Data" and "Model".
4. The best ones **label their arrows**
 - MedCPT: query / article / retrieval / re-ranking — not bare lines.
5. A pipeline can contain an architecture.
 - **Draw the flow**, then zoom into the one box that matters. They are layers, not rivals.

Today we draw the middle one.

The one rule for today

✓ Draw what you invented. Do not spend the space on what everyone already has.

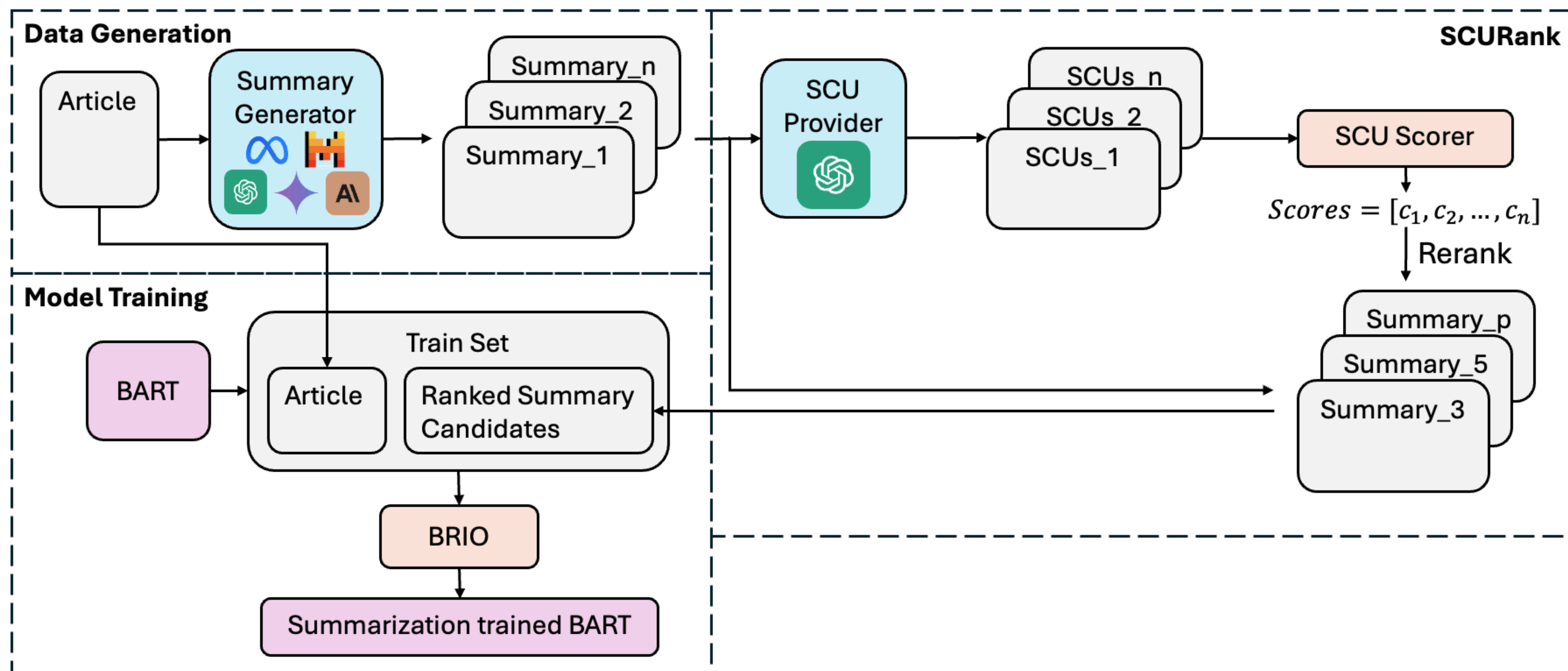
- A pre-trained encoder, an LLM, a standard optimizer → a plain box is enough.
- The step you invented → **this is where the detail goes.**

✗ the box that carries your contribution is the smallest and emptiest box in the figure.

Three rewrites

- All of these are my own figures
- *I am showing you my own bad figures so that you know everyone starts here.*

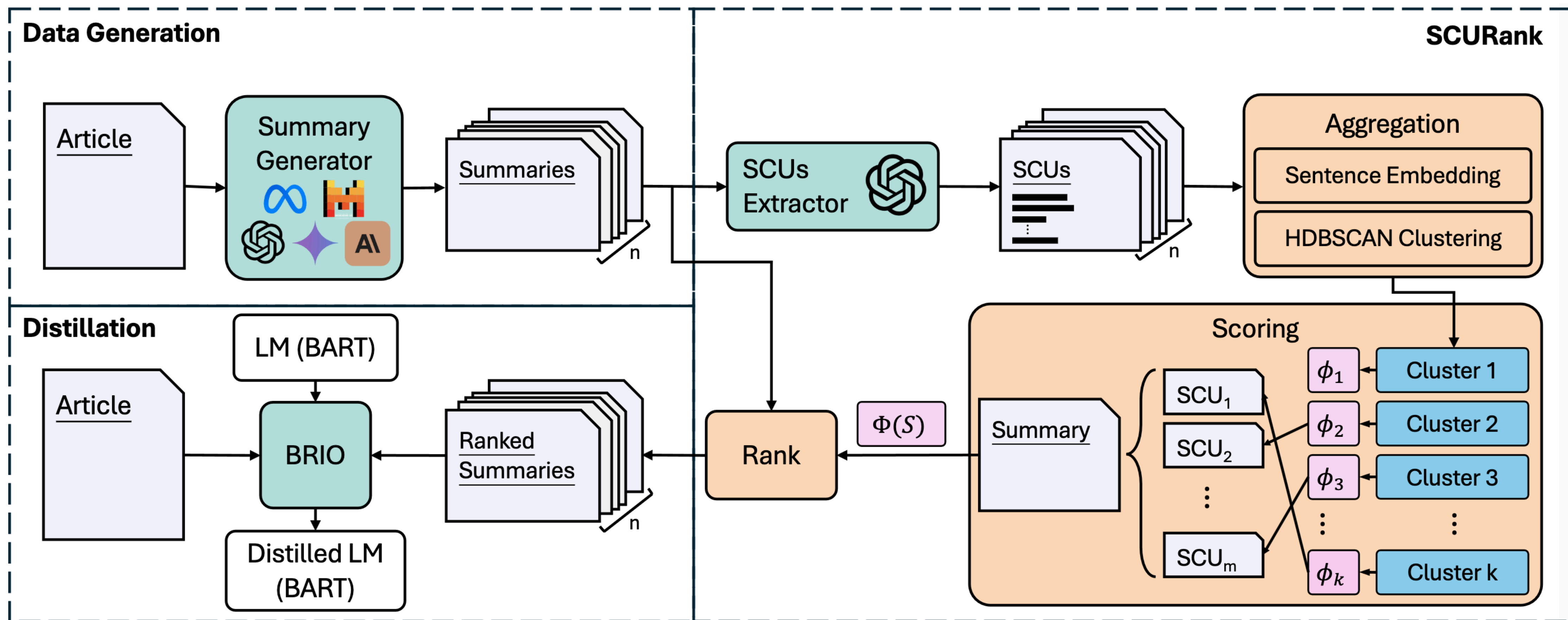
Rewrite 1 – SCURank, before



Rewrite 1 — what is wrong

1. The bottom right quarter is empty. The figure is a wide box with a hole in it.
2. SCU Scorer is one small box. It is the contribution of the paper, and it is the least detailed thing in the figure.
3. Summary_1, SCUs_n — this is code style, not paper style.
4. SCU Provider — "provider" of what? The name does not say what the step does.
5. Four colors, no meaning given. Blue, orange, pink, grey — which is which?

Rewrite 1 – SCURank, after



Rewrite 1 — what changed

Before	After
empty bottom right	same width, much more content
SCU Scorer (one box)	opened into Aggregation (Sentence Embedding → HDBSCAN) and Scoring
Summary_1, SCUs_n	stacked document icons with <i>n</i>
SCU Provider, Model Training	SCUs Extractor, Distillation
colors mean nothing	colors mean the proposed steps

Rewrite 2 – UCWS, before

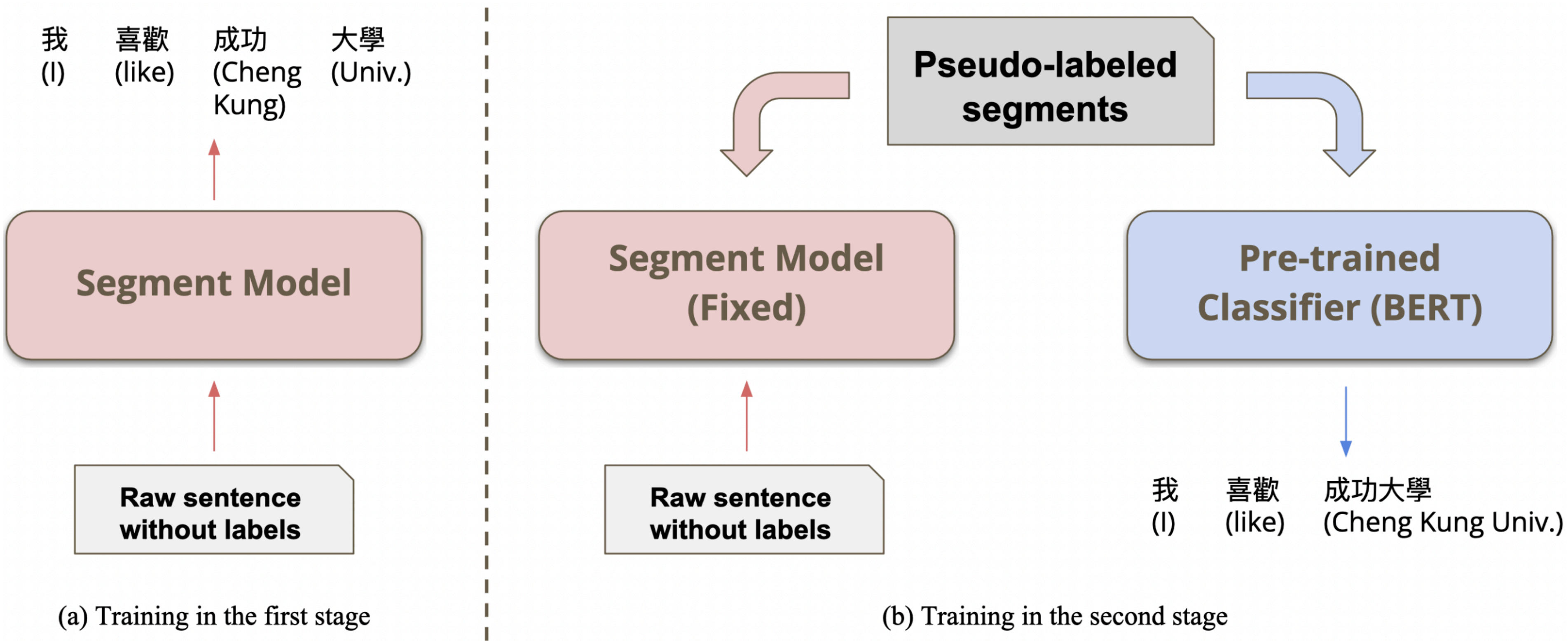


Figure 1: The proposed training framework for unsupervised Chinese word segmentation.

Rewrite 2 — UCWS, what is wrong

1. Two very large boxes that say only Segment Model. Half the figure is empty box.
2. Panel (a) is almost a copy of the left half of panel (b). The reader reads the same thing twice.
3. Two arrow styles — thin lines and thick block arrows. What is the difference? Never said.
4. The example changes between the panels — (a) 成功 / 大學 apart, (b) 成功大學 together. That change is the whole point of the method, and nothing points at it.
5. 3D edges and shadows. This is the default PowerPoint style; it adds no meaning.
6. (Fixed) is a word the reader has to notice inside a long label.

Rewrite 2 – UCWS, after

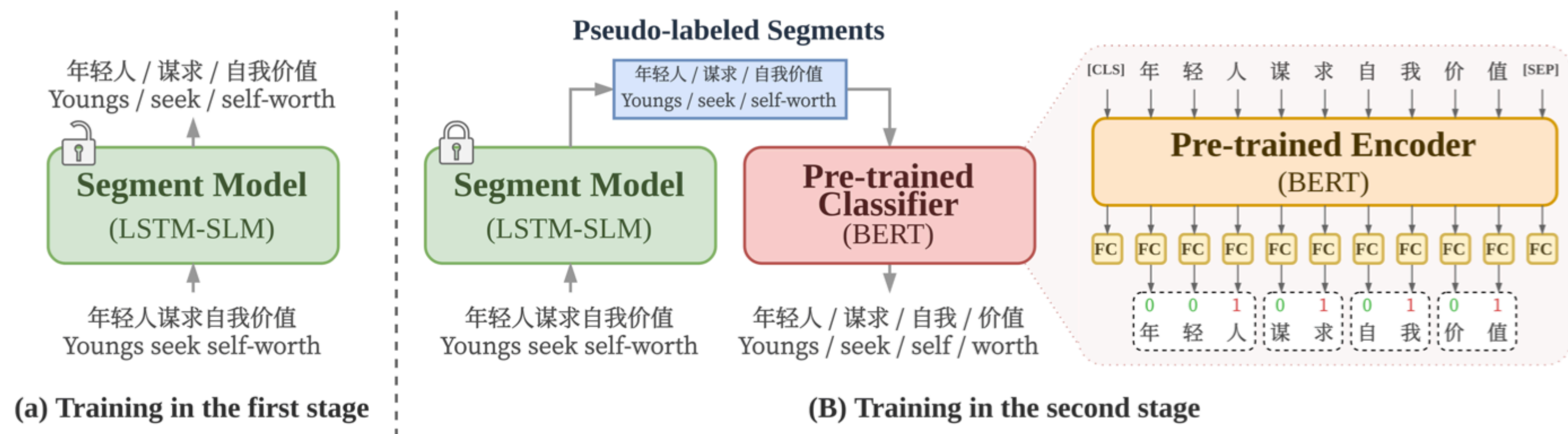


Figure 1: The proposed training framework for unsupervised Chinese word segmentation.

Rewrite 2 – UCWS, after

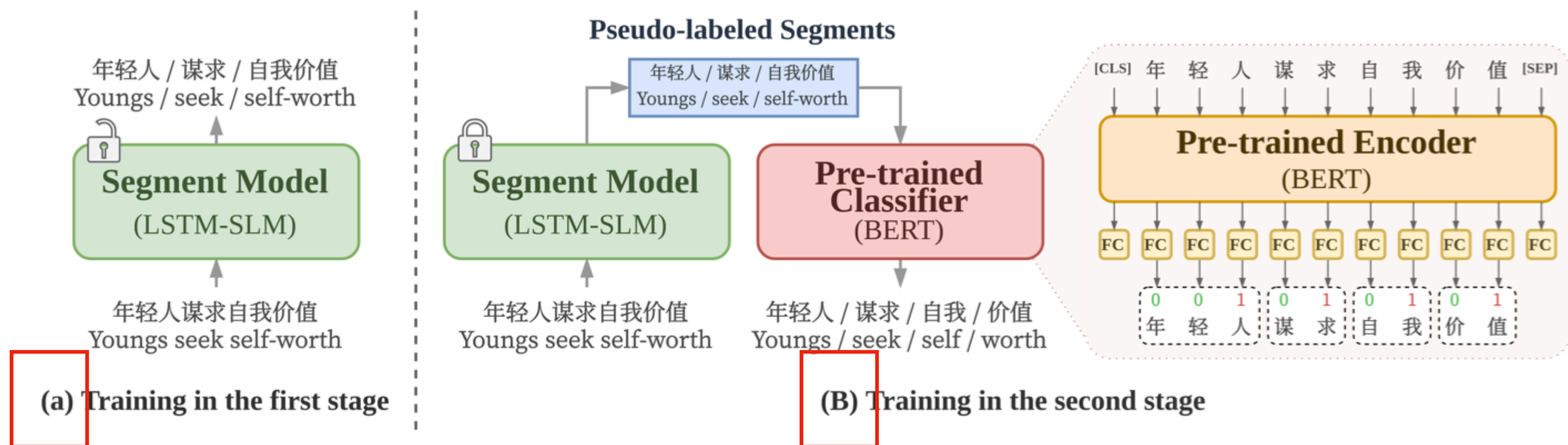
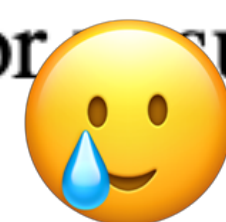


Figure 1: The proposed training framework for unsupervised Chinese word segmentation.



<https://aclanthology.org/2023.emnlp-main.564.pdf>

Rewrite 2 — what changed

Before	After
huge empty boxes	boxes fit their text — the saved space became a zoom-in panel
the classifier is a black box	the panel shows tokens → FC → 0 / 1 boundary labels
two arrow styles	one arrow style
the example changes without warning	one running example, input at the bottom, output at the top, in both panels, with the advantage brought by the proposed method
3D edges and shadows	flat (more modern)
(Fixed) as text	lock icons: open lock = trained, closed lock = frozen

Rewrite 3 — few-shot report labeling

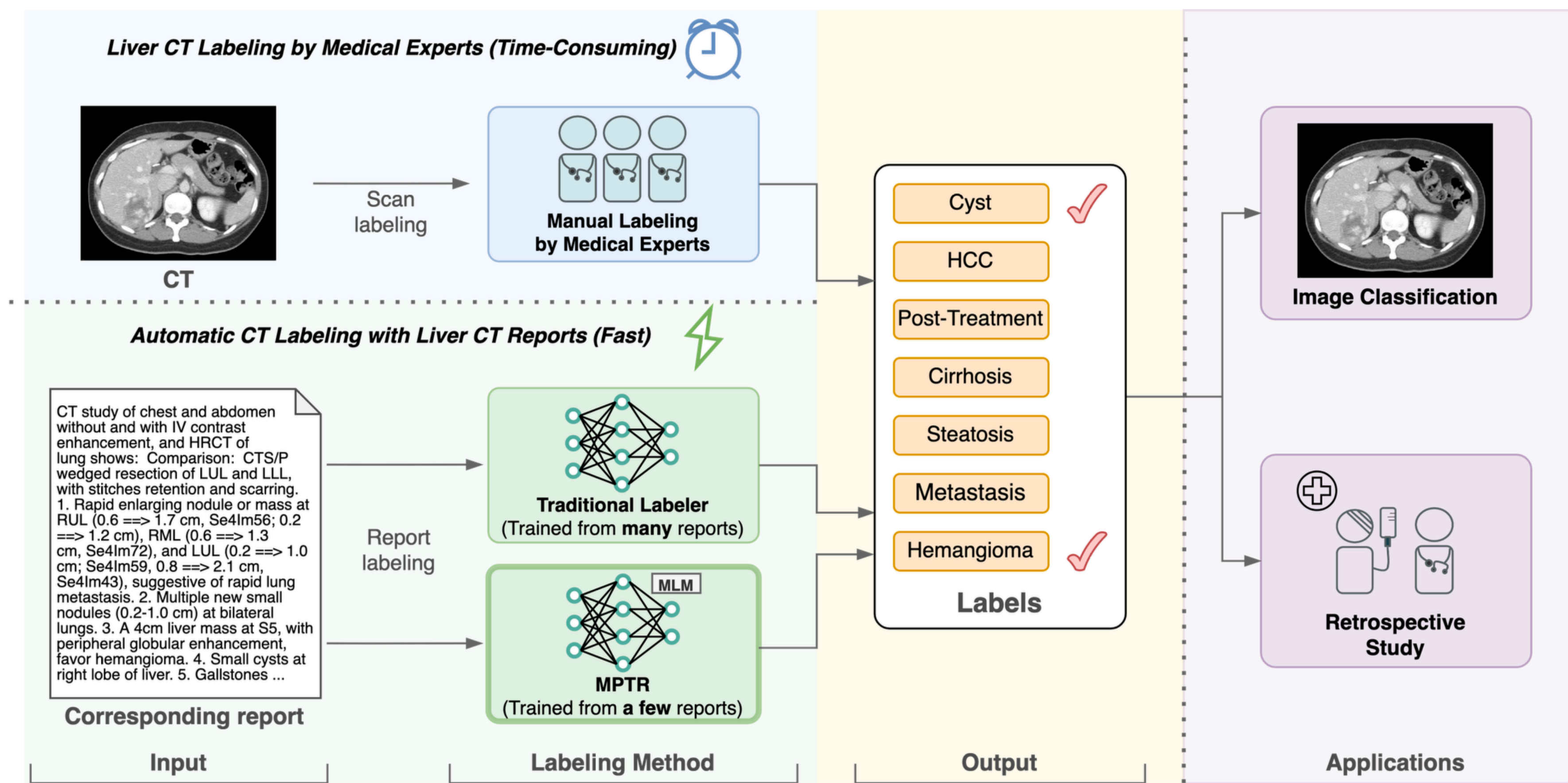


Fig. 1: The illustration of CT labeling for a comparison of manual labeling and automatic labeling through the corresponding CT reports. Real-world applications using CT scans and the labels are present at the right.

Rewrite 3 — what is wrong

1. The report is a wall of small text. Shrunk to column width, nobody can read it. It looks like information but decoration.
2. The two methods are drawn as the same picture. Traditional Labeler and MPTR are identical network drawings. The picture does not show how they differ — only the labels do.
3. The red check marks are never explained. Why Cyst and Hemangioma?
4. The CT image appears twice, unchanged and very confusing.
5. The figure never says which method is the proposed one.

BioBERT

Lee, Jinhyuk, et al. "BioBERT: a pre-trained biomedical language representation model for biomedical text mining." *Bioinformatics* 36.4 (2020): 1234-1240.

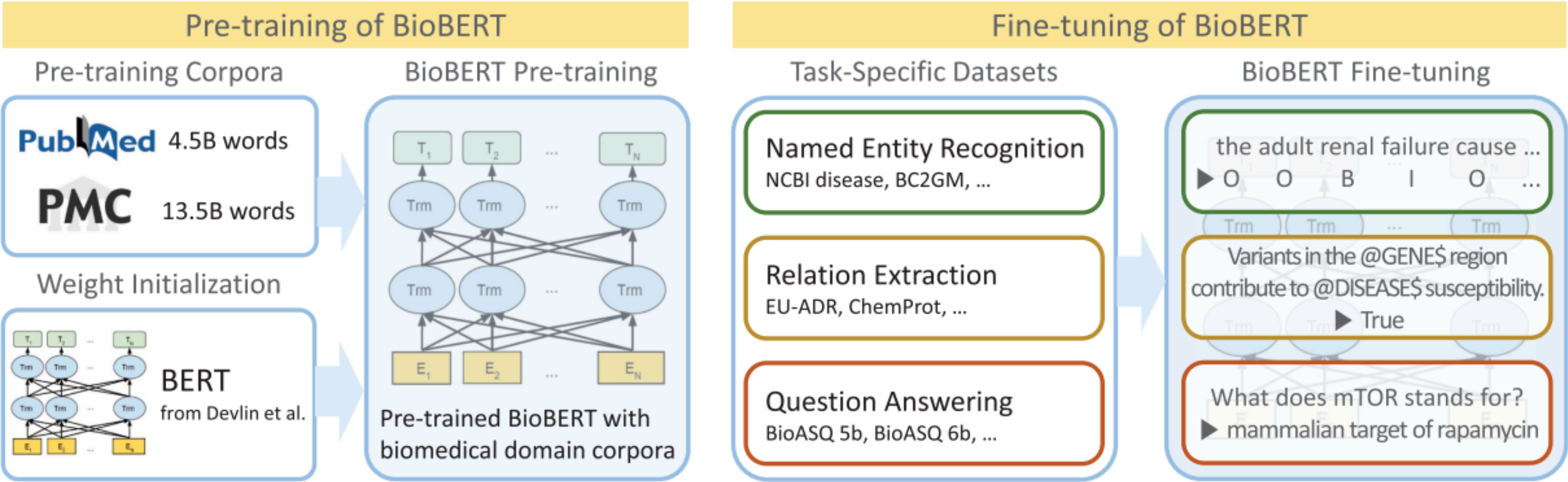


Fig. 1. Overview of the pre-training and fine-tuning of BioBERT

BioBERT — why it works

- **Two named phases with header bars:** Pre-training and Fine-tuning. You know the structure and the phases in one second.
- **Real examples** of each task — you can see O O B I O, True, mammalian target of rapamycin.
- **Color links two columns:** green = NER, yellow = relation extraction, red = QA. The colour has a job.

MedCPT

Jin, Qiao, et al. "Medcpt: Contrastive pre-trained transformers with large-scale pubmed search logs for zero-shot biomedical information retrieval." Bioinformatics 39.11 (2023): btad651.

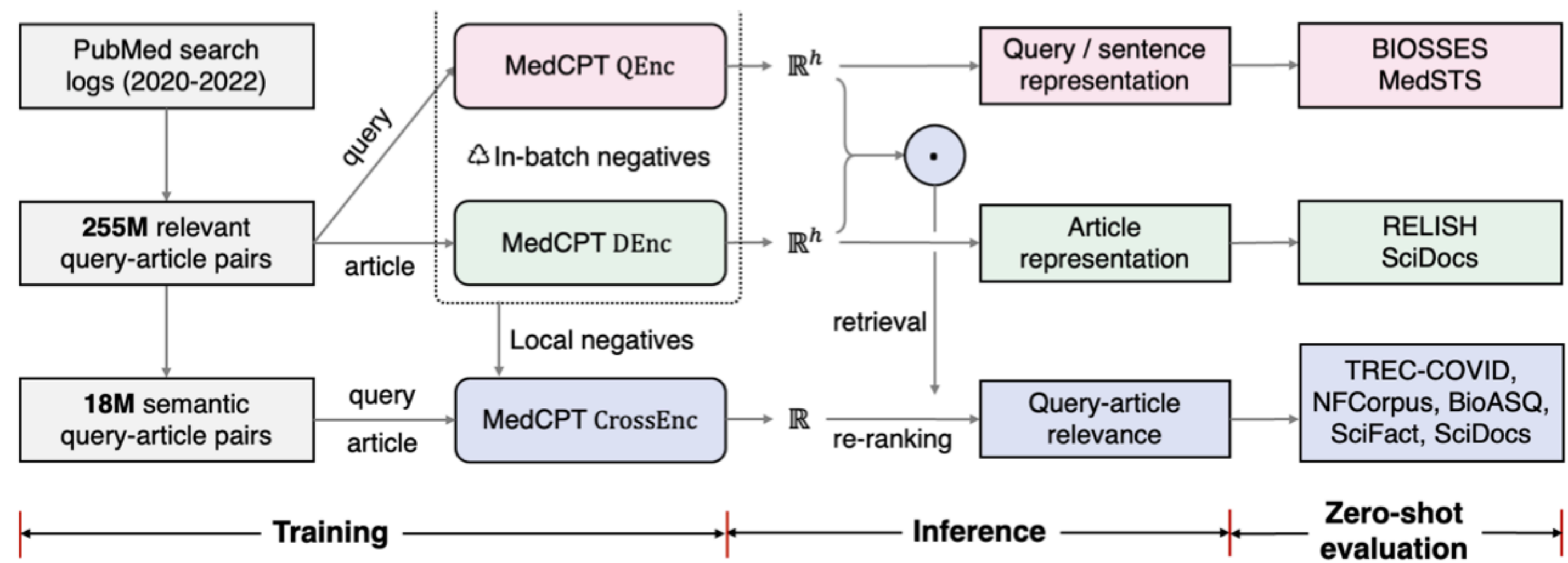


Figure 1. A high-level overview of this work. MedCPT contains a query encoder (QEnc), a document encoder (DEnc), and a cross-encoder (CrossEnc). The query encoder and document encoder compose of the MedCPT retriever, which is contrastively trained by 255M query-article pairs and in-batch negatives from PubMed logs. The cross-encoder is the MedCPT re-ranker, and is contrastively trained by 18M non-keyword query-article pairs and local negatives retrieved from the MedCPT retriever. MedCPT achieves state-of-the-art performance on various biomedical information retrieval tasks under zero-shot settings, including query-article retrieval, sentence representation, and article representation.

MedCPT — why it works

- 💡 **Every arrow is labelled:** query, article, retrieval, re-ranking, Local negatives.
Compare this to my figures, where the arrows say nothing.
- A reading axis at the bottom: Training | Inference | Zero-shot evaluation.
- Color follows a component from left to right — pink QEnc → pink representation → pink benchmarks.
- Real names everywhere — 255M pairs, 18M pairs, and the actual benchmark names.

The drawing checklist

Before you open your laptop, know what you are aiming for:

- A reader can trace **input** → **output** in five seconds
- **The step you invented** is the most detailed part
- Every arrow is labelled, or **all arrows clearly mean one thing**
- Color means something, and the meaning is written down
- Every symbol in the figure also appears in the text, spelled the same way
- **No text smaller** than the caption font
- Nothing appears twice
- It still works printed in black and white

Your Turn

NEXT WEEK ATTENTION!!

- Next week we will write the **Method** section.
- Please prepare your method sketch (plain text or text in items).

Thank you for your participation 